VS_helper

Local Code Completion for VS Code

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DA Final Project Demo
Individual submission

Problem Statement and Scope

- Cloud code completion needs an external service and internet access.
- The aim is to show code suggestions using a model on the same computer.
- The extension supports Python, JavaScript, TypeScript, JSX and TSX.
- It uses nearby code in the active file. Full repository search is outside the current scope.

System Architecture and Design



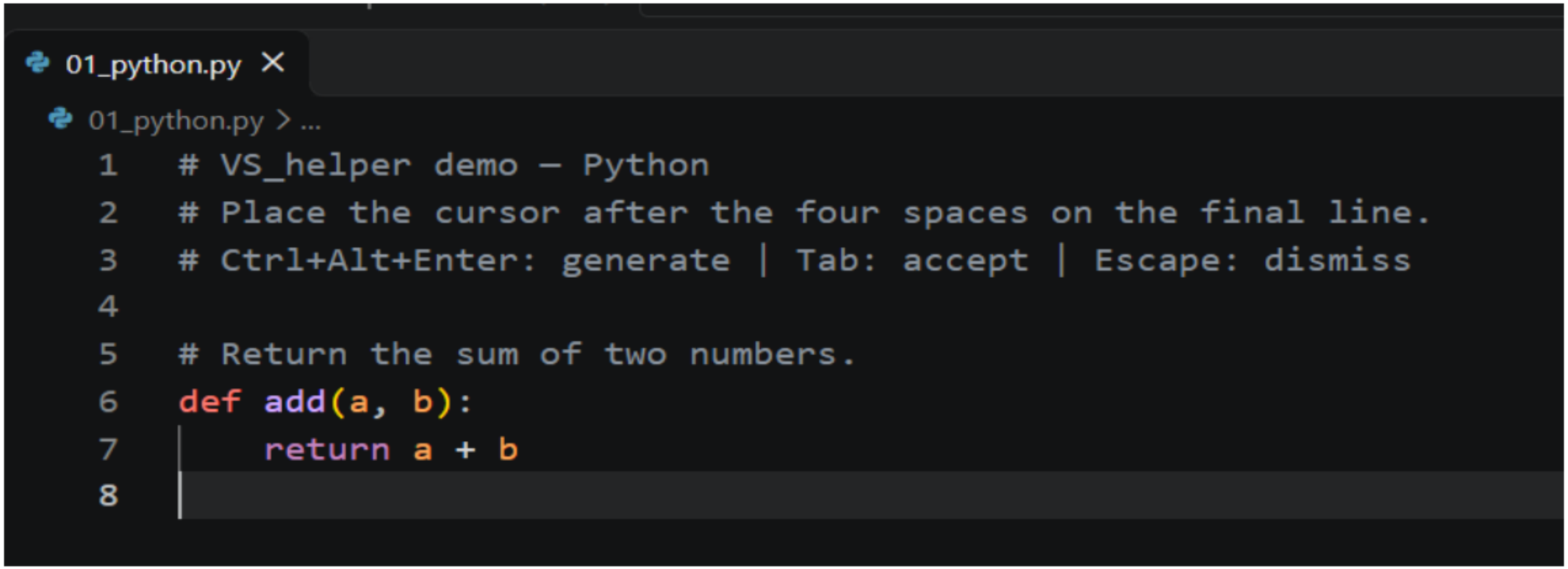
TypeScript extension + VS Code API + Ollama + Qwen 2.5 Coder 1.5B

Local endpoint: http://127.0.0.1:11434
The developer reviews the suggestion and presses Tab to insert it.

Key Features

- Native ghost-text suggestions with Tab to accept and Escape to dismiss.
- Model selection, connection testing and an enable or disable control.
- A short typing delay and cancellation of outdated requests.
- Context limits and cleanup of repeated code or extra explanations.
- Loopback-only Ollama addresses, with HTTP redirects blocked.

Live Demo Screenshot



Python example: the accepted suggestion completes the add function.

Testing and Current Result

- 43 unit tests passed for context, cleanup, cancellation and transport behavior.
- Local Python, JavaScript and TypeScript examples passed simple function checks.
- The live editor test passed native suggestion triggering and acceptance.
- The current demo runs qwen2.5-coder:1.5b through local Ollama.
- These small tests do not guarantee that every generated suggestion is correct.

Challenges and Future Improvements

Challenges

Slow and interrupted model download
Extra code after the requested function
Shortcut conflicts with other apps

Future improvements

Test larger and more varied coding tasks
Compare more local code models
Measure real editing delay and suggestion acceptance

Source Code and Local Setup

Public repository

https://github.com/Soorya-Ray/VS_helper

- Install Node.js, VS Code and Ollama, then pull qwen2.5-coder:1.5b.
- Run npm install and npm run package. Install the generated VSIX.
- Select Ollama and test the provider connection in VS_helper.
- Open a demo file, request a suggestion and accept it with Tab.

Live app URL: not applicable for this desktop extension.